Aga Khan University Examination Board

Notes from E-Marking Centre on HSSC-II Business Statistics Examination May 2018

Introduction:

This document has been produced for the teachers and candidates of Higher Secondary School Certificate (HSSC-II) Business Statistics. It contains comments on candidates' responses to the 2018 HSSC-II Examination, indicating the quality of the responses and highlighting their relative strengths and weaknesses.

E- Marking Notes:

This includes overall comments on candidates' performance on every question and *some* specific examples of candidates' responses which support the mentioned comments. Please note that the descriptive comments represent an overall perception of the better and weaker responses as gathered from the e-marking session. However, the candidates' responses shared in this document represent some specific example(s) of the mentioned comments.

Teachers and candidates should be aware that examiners may ask questions that address the Student Learning Outcomes (SLOs) in a manner that require candidates to respond by integrating knowledge, understanding and application skills they have developed during the course of study. Candidates are advised to read and comprehend each question carefully before writing the response to fulfil the demand of the question.

Candidates need to be aware that the marks allocated to the questions are related to the answer space provided on the examination paper as a guide to the length of the required response. A longer response will not in itself lead to higher marks. Candidates need to be familiar with the command words in the SLOs which contain terms commonly used in examination questions. However, candidates should also be aware that not all questions will start with or contain one of the command words. Words such as 'how', 'why' or 'what' may also be used.

Key observations:

This year candidates did not perform well in questions based on price index, permutation and combination were not well attempted. Weaker responses revealed that candidates had problems with finding median and mode for grouped data and probability.

Detailed Comments:

Constructed Response Questions (CRQs)

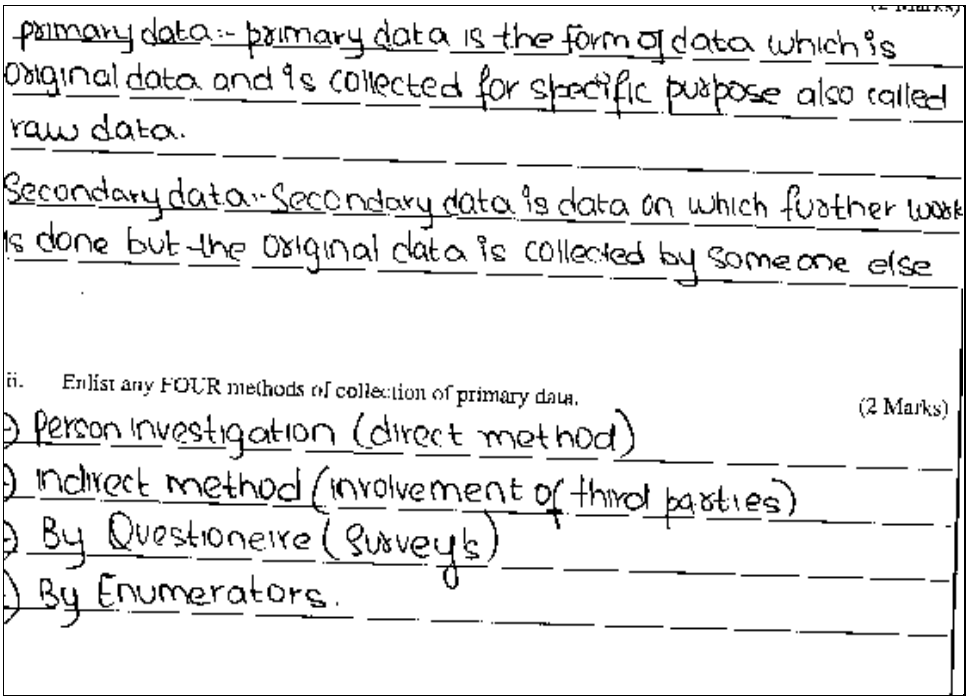
Question 1:

i. Define primary and secondary data.

ii. Enlist any FOUR methods of collection of primary data.

Better responses indicated that in part **i**, the candidates were able to define primary and secondary data correctly. In part **ii**, candidates enlisted correct methods of collection of primary data, as given in the following example.

Example:


primary data:- primary data is the form of data which is original data and is collected for specific purpose also called raw data.

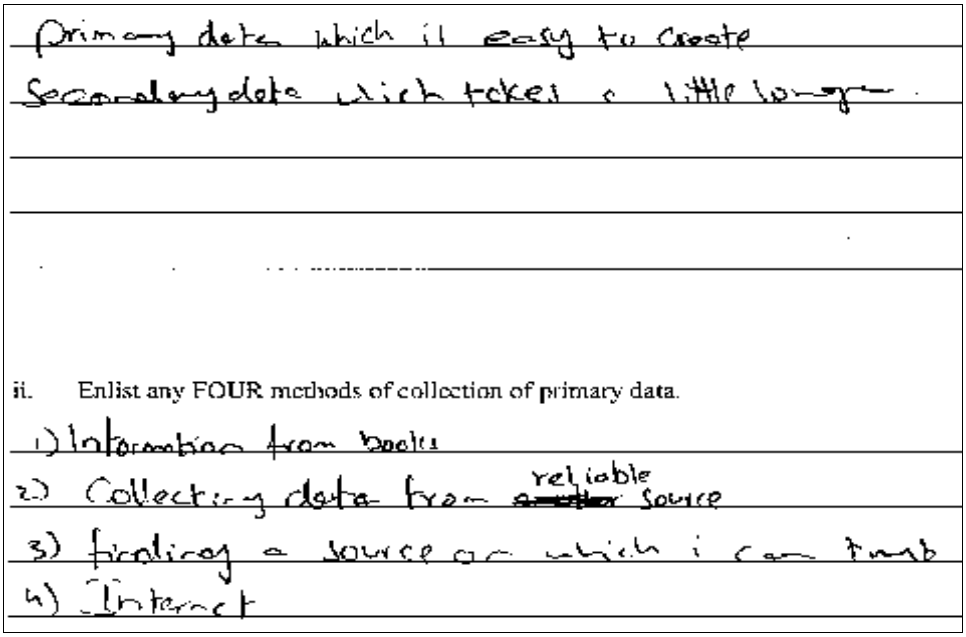
Secondary data:- Secondary data is data on which further work is done but the original data is collected by someone else

ii. Enlist any FOUR methods of collection of primary data. (2 Marks)

) Person investigation (direct method)
) Indirect method (involvement of third parties)
) By Questioneire (Surveys)
) By Enumerators.

Weaker responses revealed that in part i, candidates lacked knowledge of primary and secondary data and failed to define the terms asked in the question properly. The responses of part ii reflected that candidates had no idea about the method of collection of primary data and wrote random responses as cited in the given example.

Example:


Primary data which is easy to create
Secondary data which takes a little longer.

ii. Enlist any FOUR methods of collection of primary data.

1) Information from books
2) Collecting data from ~~another~~ reliable source
3) finding a source on which i can trust
4) Internet

Question 2:

The given frequency distribution represents the weight (in kilograms) of 50 employees at an organisation.

Weight (kg)	Number of Employees
40 – 49	3
50 – 59	11
60 – 69	17
70 – 79	14
80 – 89	5

Find the

i. class width of the given frequency distribution.

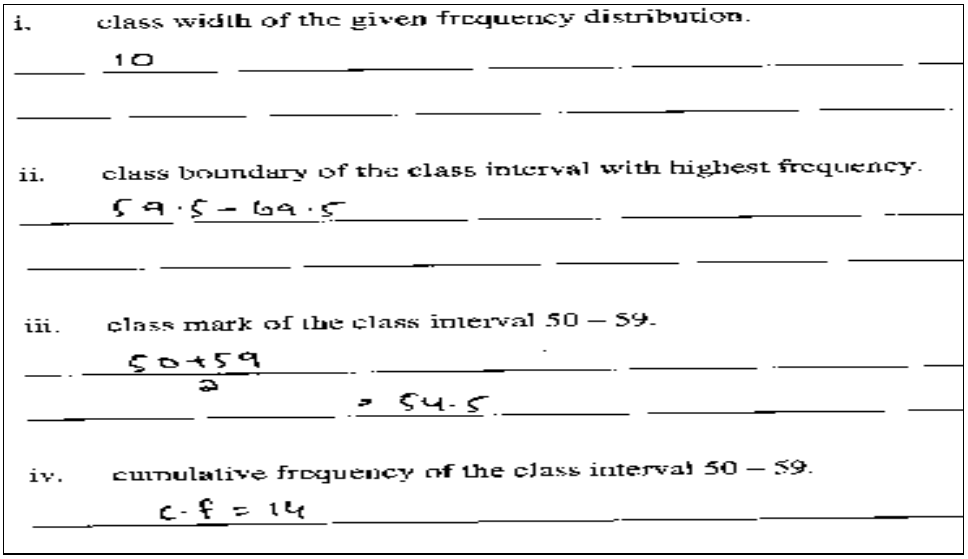
ii. class boundary of the class interval with the highest frequency.

iii. class mark of the class interval 50 – 59.

iv. cumulative frequency of the class interval 50 – 59.

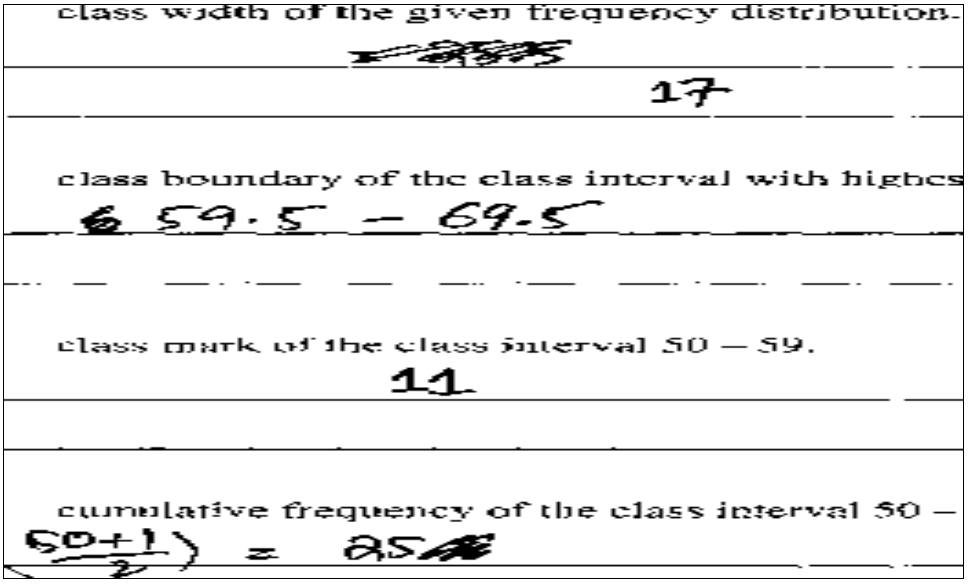
Better responses showed that the candidates were well aware of the concepts asked in the question related to grouped data. They found the class width, class boundary, class mark and required cumulative frequency aptly.

Example:



Weaker responses displayed that candidates lacked in the concepts being tested in the question and failed to find the required quantities related to the frequency distribution. They used wrong formulae to find the required quantities, although formulae were fairly simple.

Example:



Question 3:

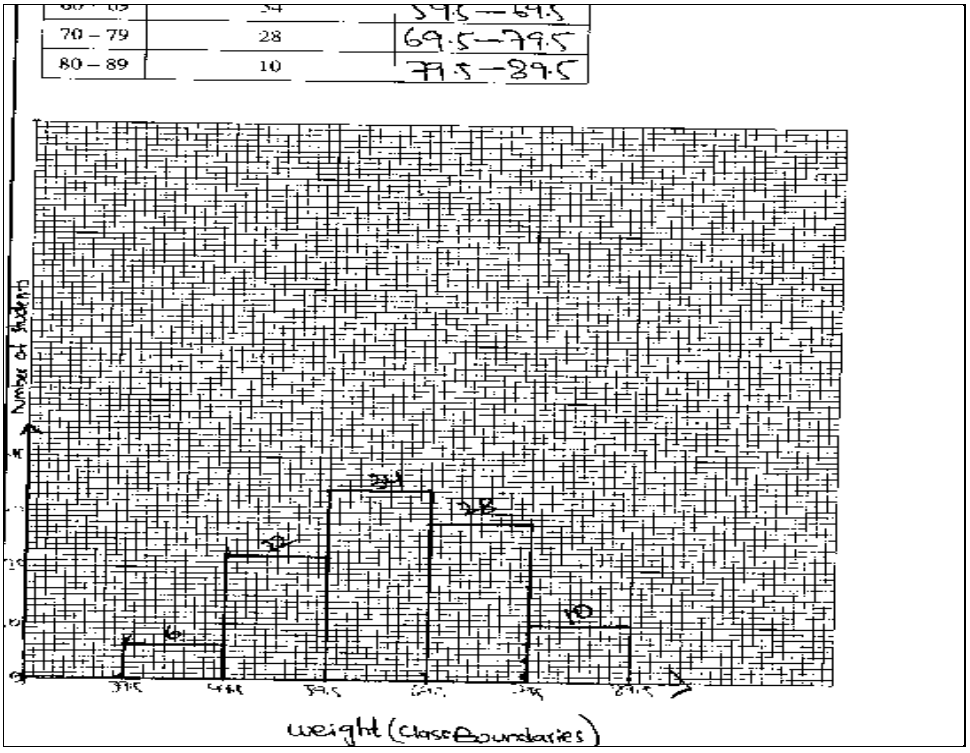
The given frequency distribution represents the weight (in kilograms) of 100 students in a school.

Find the class boundaries of each class interval and represent the given data using a histogram.

Weight (kg)	Number of Students	Class Boundary
40 – 49	6	
50 – 59	22	
60 – 69	34	
70 – 79	28	
80 – 89	10	

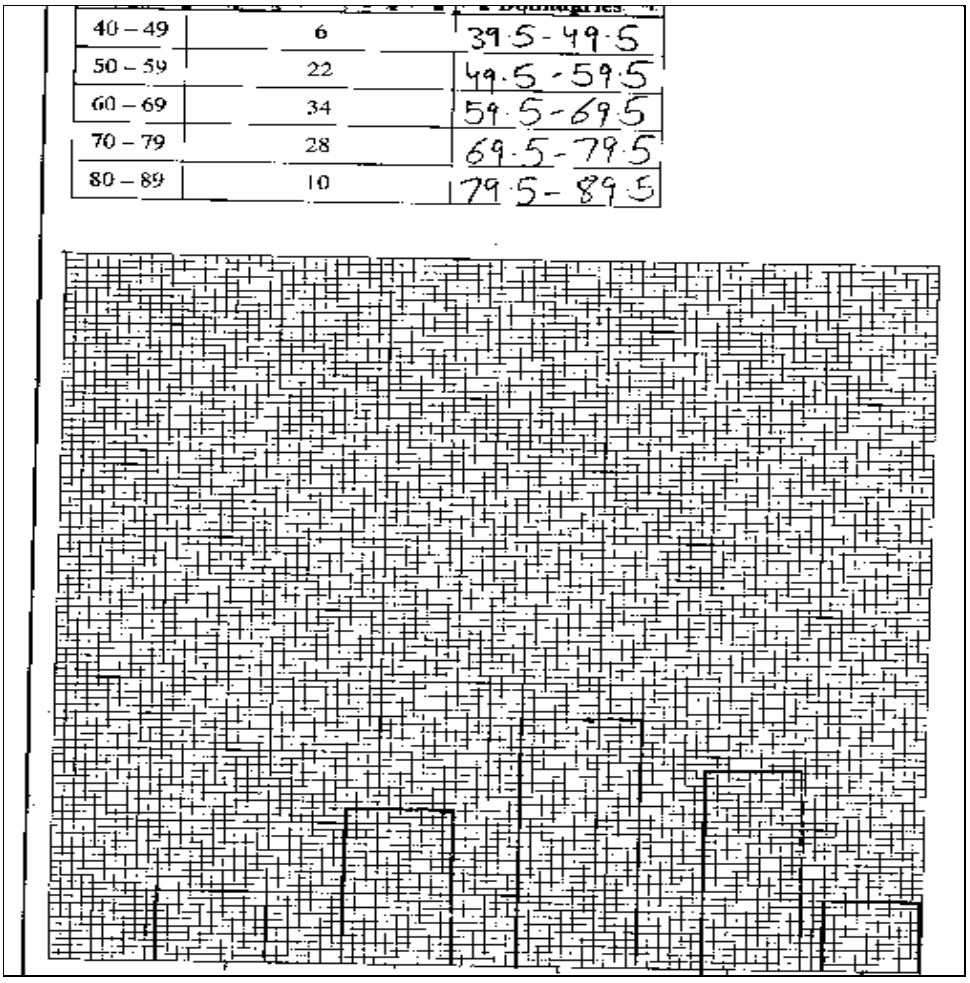
Better responses reflected that candidates were able to find the class boundaries to draw the histogram. The candidates took appropriate scale on both the axes and clearly labelled the axes to represent the data through histogram effectually.

Example:



Weaker responses indicated that candidates constructed histogram incorrectly. They failed to label the axes and unable to choose the proper scale to construct the required histogram for precise representation of data.

Example 1:



Question 4a:

The weights (in kilograms) of 150 members of a health club, who are following a diet plan, are represented by the given frequency distribution.

i. Complete the given table to find the mean weight of the club members.

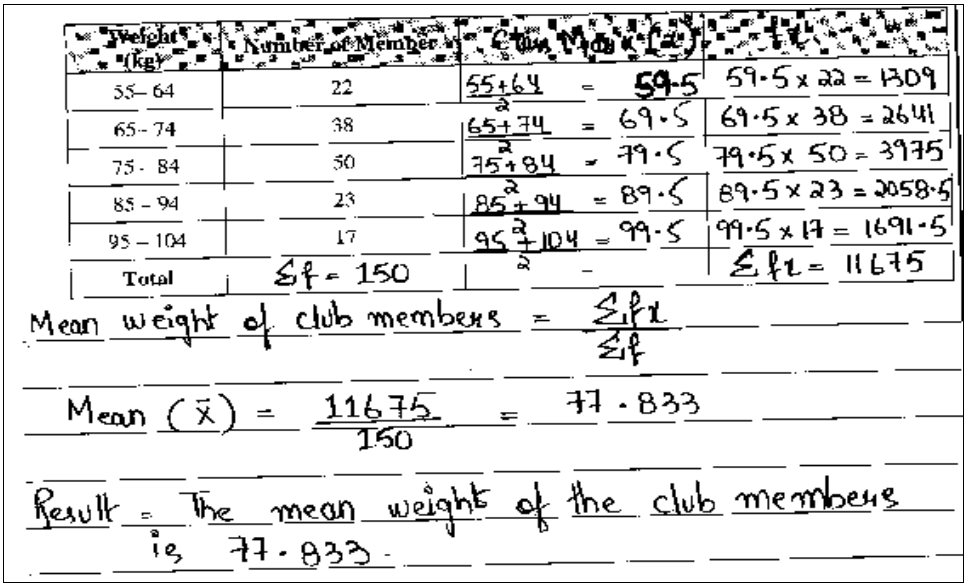
Weight (kg)	**Number of Members**		
55– 64	22		
65– 74	38		
75 – 84	50		
85 – 94	23		
95 – 104	17		
Total		**–**	

ii. If each member of the club reduces 2 kg weight, find the mean of the reduced weights using your answer of part i.

This was generally a well attempted question.

Better responses reflected that candidates found class mark of each class interval and then multiplied each class mark with its corresponding frequency. Subsequently they correctly calculated the sum $\sum f \times x$ and $\sum f$ and applied formula of arithmetic mean to find the required mean. They were well-versed with the properties of arithmetic mean to find the reduced weight, as required in part **ii** of the question.

Example



Weight (kg)	Number of Member	Class Mark (x)	fx
55 – 64	22	$\frac{55+64}{2} = 59.5$	$59.5 \times 22 = 1309$
65 – 74	38	$\frac{65+74}{2} = 69.5$	$69.5 \times 38 = 2641$
75 – 84	50	$\frac{75+84}{2} = 79.5$	$79.5 \times 50 = 3975$
85 – 94	23	$\frac{85+94}{2} = 89.5$	$89.5 \times 23 = 2058.5$
95 – 104	17	$\frac{95+104}{2} = 99.5$	$99.5 \times 17 = 1691.5$
Total	$\Sigma f = 150$		$\Sigma fx = 11675$

Mean weight of club members $= \frac{\Sigma fx}{\Sigma f}$

Mean $(\bar{x}) = \frac{11675}{150} = 77.833$

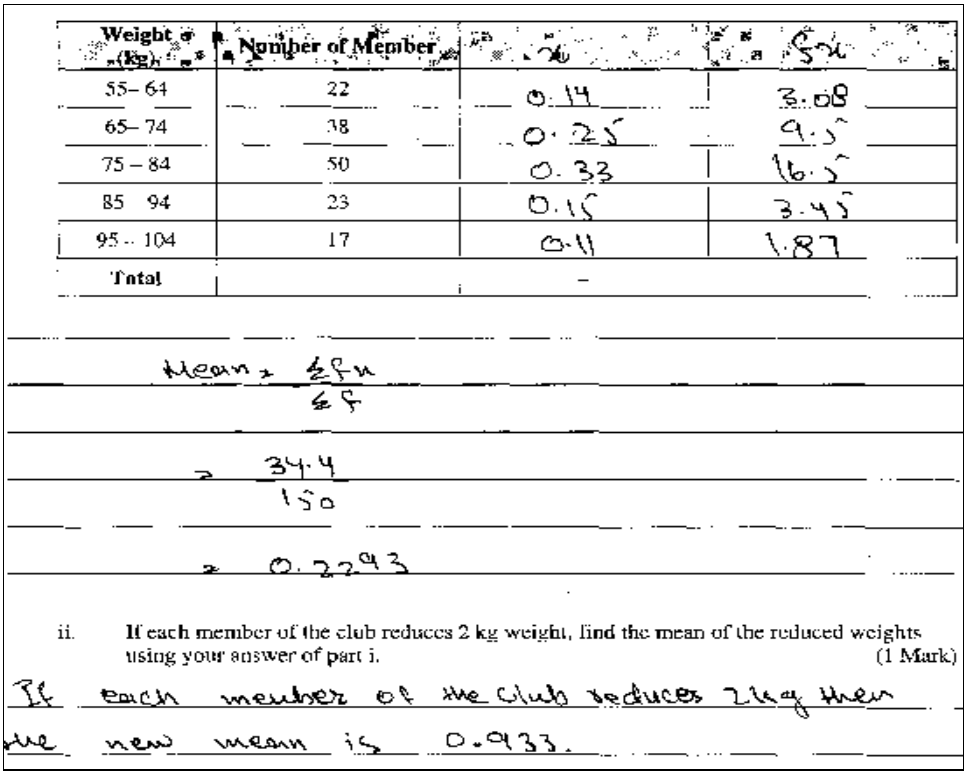
Result = The mean weight of the club members is 77.833.

Weaker responses revealed that the candidates made different types of mistakes which are reflective of their misconception or lack of understanding of arithmetic mean of grouped data. Few of the mistakes noted are as follows:

- They were unable to find the class mark correctly.
- They made mistakes in multiplication of the class marks with their corresponding frequencies.
- They made mistakes in finding the sums required in the formula of arithmetic mean.
- They failed to apply the formula correctly.
- They failed to apply the property of arithmetic mean to find the reduced weight.

Two examples are given to cite these kind of mistakes.

Example 1:



Weight (kg)	Number of Member	x	fx
55– 64	22	0.14	3.08
65– 74	38	0.25	9.5
75 – 84	50	0.33	16.5
85 94	23	0.15	3.45
95 - 104	17	0.11	1.87
Total		–	

Mean = $\frac{\Sigma fx}{\Sigma f}$

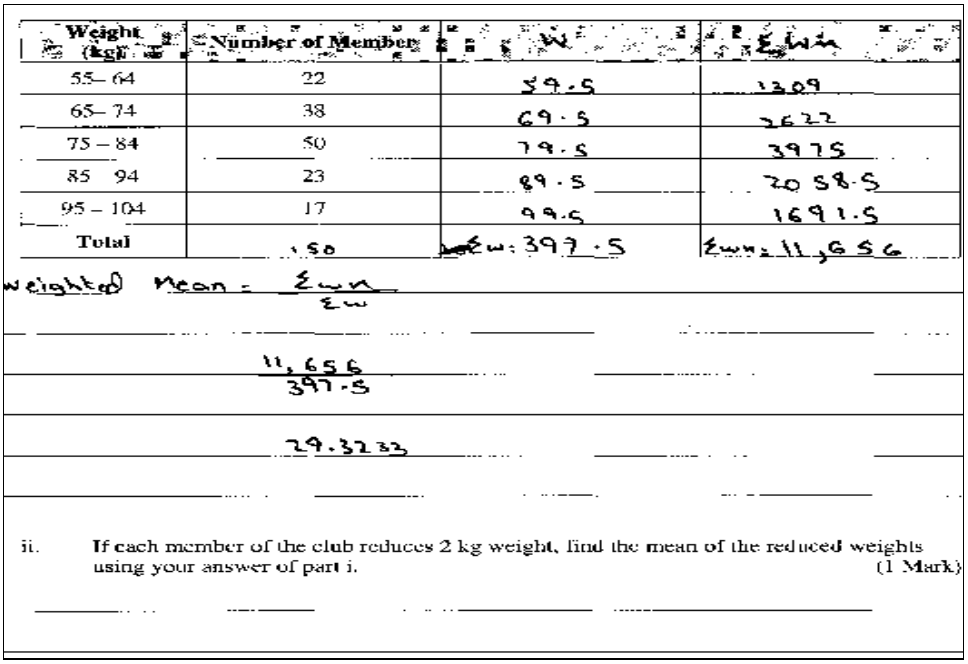
= $\frac{34.4}{150}$

= 0.2293

ii. If each member of the club reduces 2 kg weight, find the mean of the reduced weights using your answer of part i. (1 Mark)

If each member of the club reduces 2kg then the new mean is 0.093.

Example 2:



Weight (kg)	Number of Member	w	Σwm
55– 64	22	59.5	1309
65– 74	38	69.5	2622
75 – 84	50	79.5	3975
85 94	23	89.5	2058.5
95 – 104	17	99.5	1691.5
Total	150	Σw: 397.5	Σwm: 11,656

weighted mean = $\frac{\Sigma wm}{\Sigma w}$

$\frac{11,656}{397.5}$

29.3233

ii. If each member of the club reduces 2 kg weight, find the mean of the reduced weights using your answer of part i. (1 Mark)

Question 4b:

The following are the sales (in Rupees) of tomatoes in twenty days.

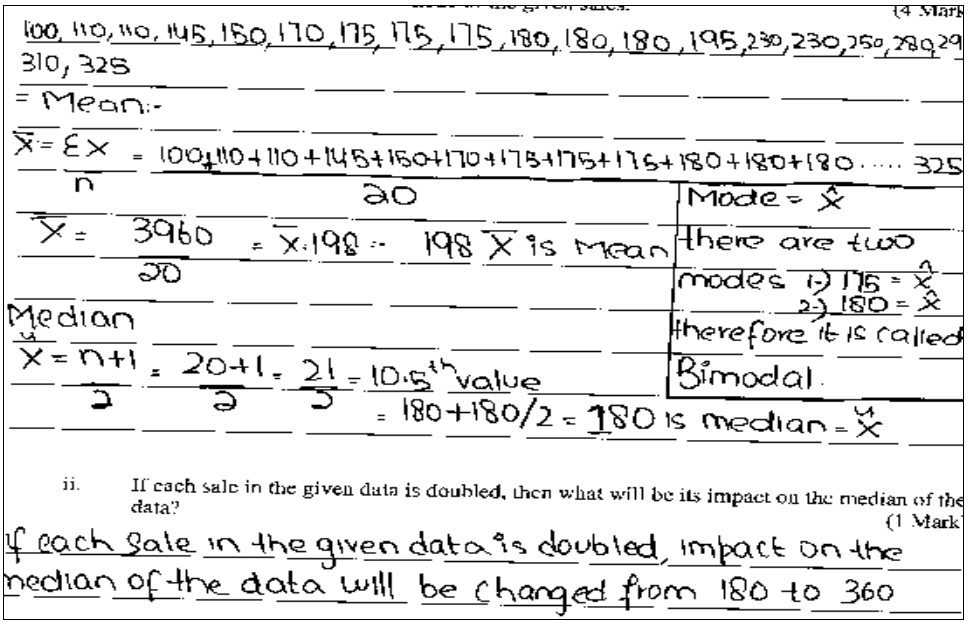
310	325	110	175
290	110	150	280
230	170	100	195
180	180	175	145
230	250	175	180

i. Find the mean, median and mode of the given sales.

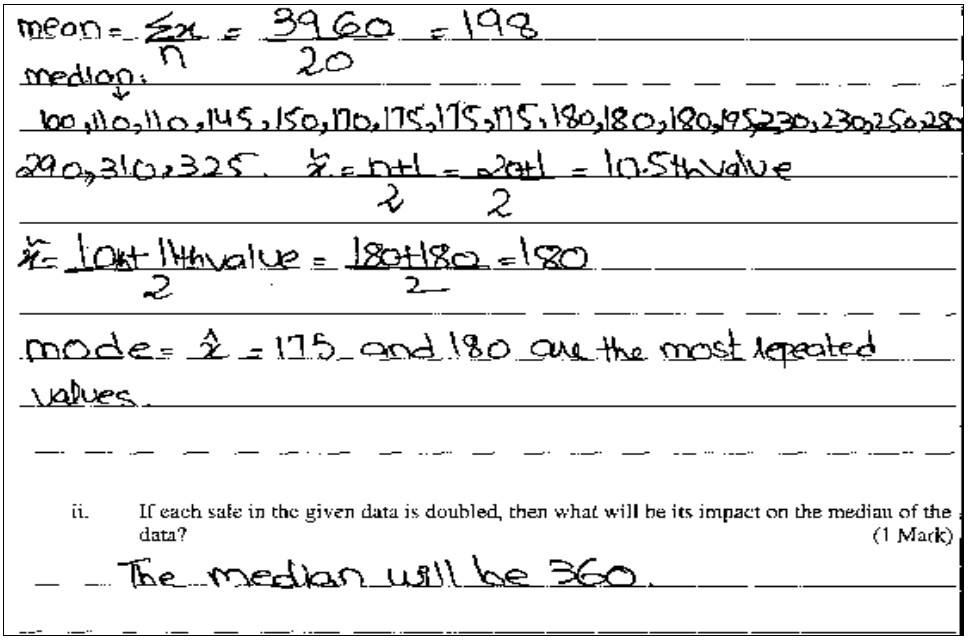
ii. If each sale in the given data is doubled, then what will be its impact on the median of the data?

Better responses reported that the candidates were able to find the mean, median and mode of the given ungroup data. They arranged the data in ascending or descending order to find the median and mode of the data. They added all the values given in the data to find the arithmetic mean by using formula correctly. Also, they understood and found the impact on median after doubling of each value of the data.

Example 1:

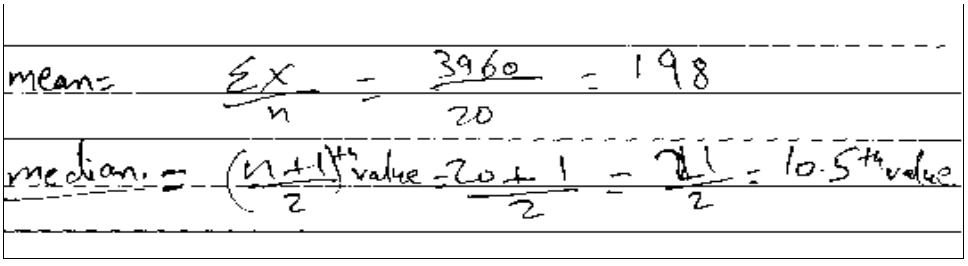


Example 2:

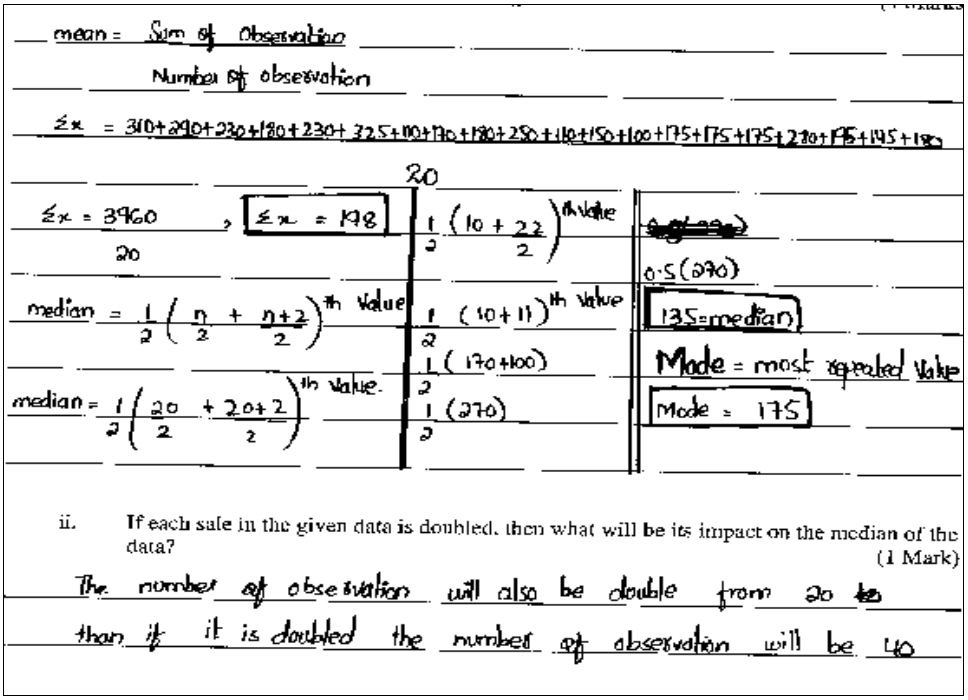


Weaker responses showed that the candidates were able to find the arithmetic mean correctly but showed different types of confusions while finding the mode and median of the given data. It was a bimodal data but they found either single mode or both modes incorrectly. Similarly they were failed to decide that which terms of the data is representing median of the given data. The weaker responses indicated that candidates were failed to write the impact on median if each item of the data is doubled.

Example 1:



Example 2:



Question 5:

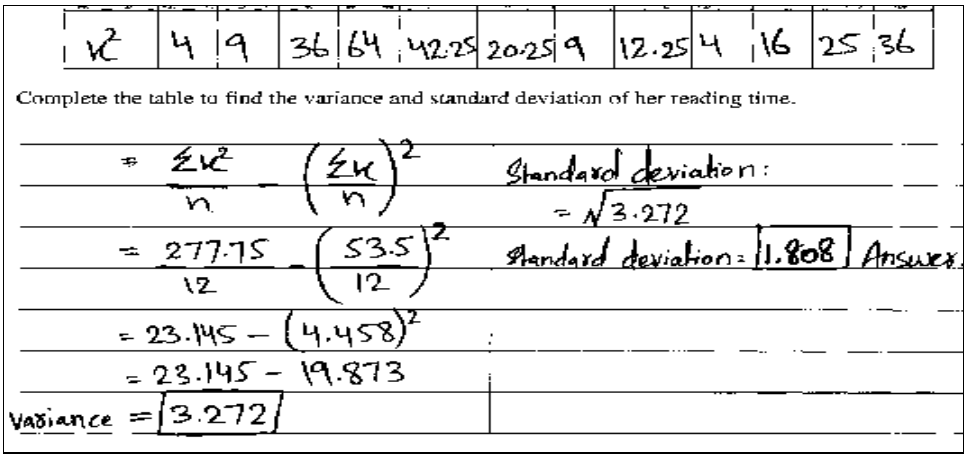
Samina is fond of reading books. She noted her reading time for the first 12 days of a month. The time noted (in hours) is as follows:

Time (hours)	2	3	6	8	6.5	4.5	3	3.5	2	4	5	6

Complete the table to find the variance and standard deviation of her reading time.

Better responses indicated that the candidates were able to understand the given scenario and apply their skills in finding the standard deviation as per the given data. They correctly found the values of $\sum x$ and $\sum x^2$ and then, substituted these values in the formula $\sqrt{\frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2}$ to find the required standard variance of the data.

Example:



x^2	4	9	36	64	42.25	20.25	9	12.25	4	16	25	36

Complete the table to find the variance and standard deviation of her reading time.

$= \frac{\Sigma x^2}{n} - \left(\frac{\Sigma x}{n}\right)^2$

$= \frac{277.75}{12} - \left(\frac{53.5}{12}\right)^2$

$= 23.145 - (4.458)^2$

$= 23.145 - 19.873$

Variance $= \boxed{3.272}$

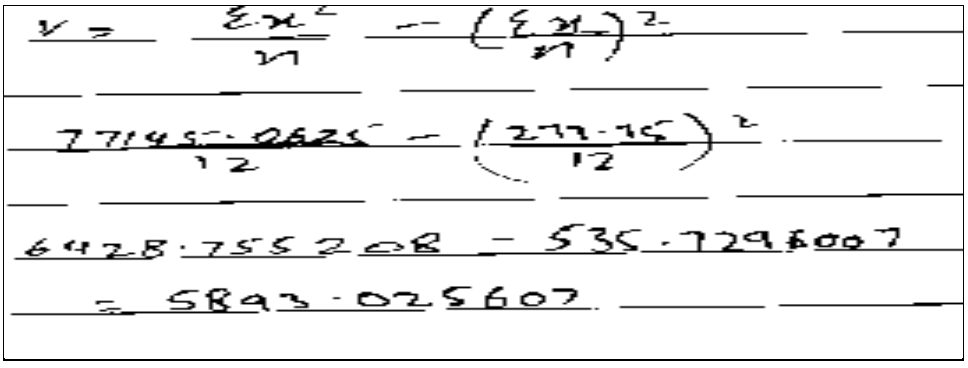
Standard deviation:

$= \sqrt{3.272}$

Standard deviation $= \boxed{1.808}$ Answer.

Weaker responses indicated that most of the candidates wrote the correct formula of the standard deviation but made mistakes in calculation of the values of $\sum x$ and $\sum x^2$. Moreover, they were unable to substitute values in the formula accurately.

Example 1:



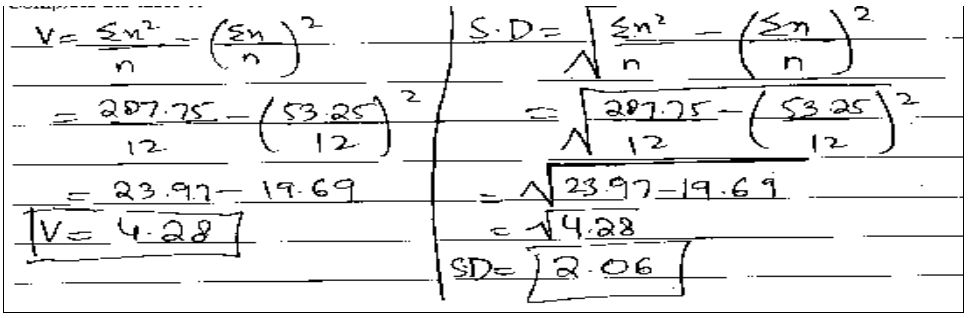
$v = \frac{\Sigma x^2}{n} - \left(\frac{\Sigma x}{n}\right)^2$

$\frac{7714.5625}{12} - \left(\frac{277.75}{12}\right)^2$

$642.8.755208 - 535.7296007$

$= 5893.025607$

Example 2:



Question 6ai:

Define the following terms.

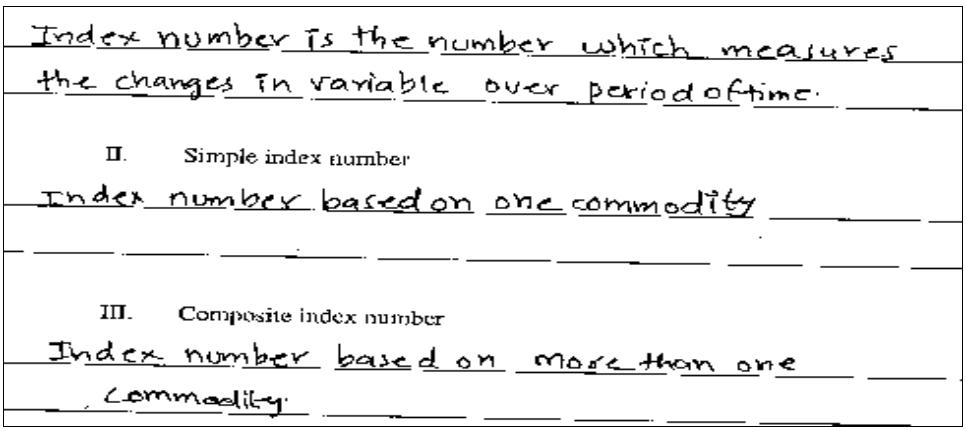
I. Index number

II. Simple index number

III. Composite index number

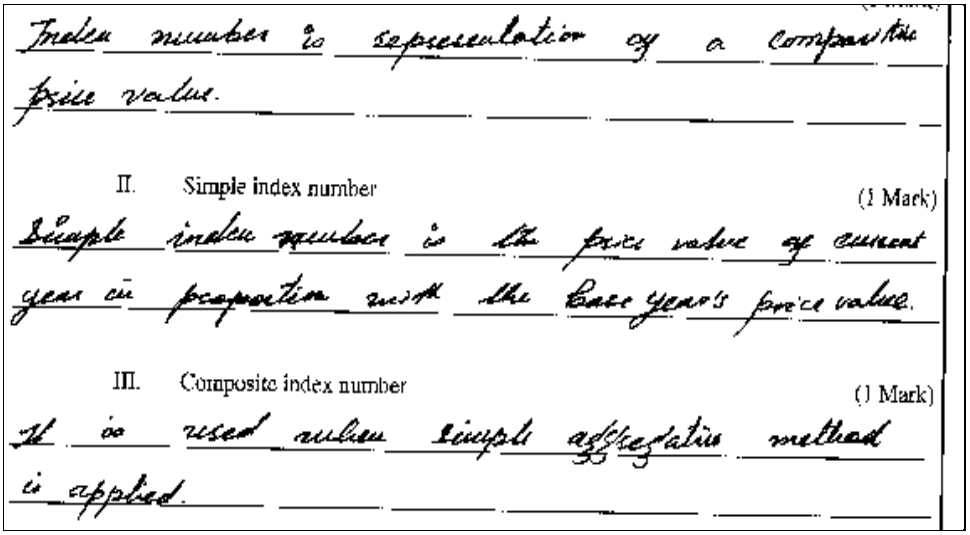
Better responses wrote the correct definition of the required terms. They had clearity about the terms asked in the question.

Example:



Weaker responses displayed that candidates did not know the proper definition of the terms and wrote unnecessary and irrelevant responses.

Example 1:


Index number is representation of a composite price value.

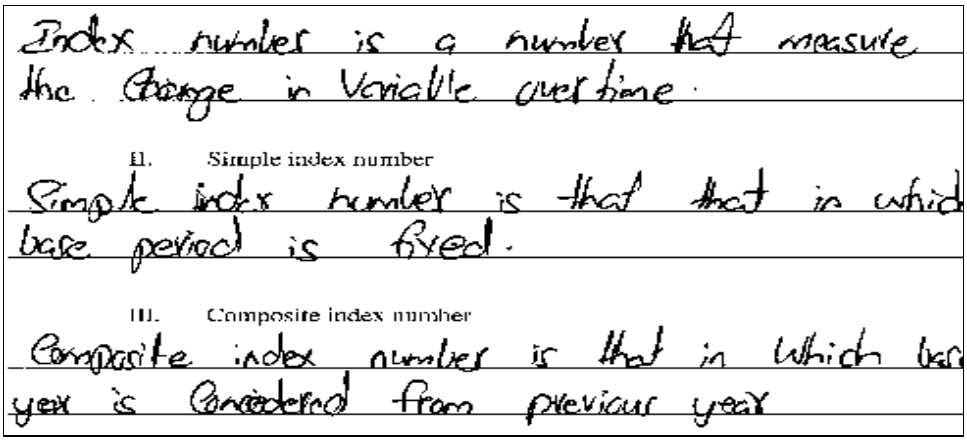
II. Simple index number (1 Mark)

Simple index number is the price value of current year in proportion with the base year's price value.

III. Composite index number (1 Mark)

It is used when simple aggregation method is applied.

Example 2:


Index number is a number that measure the change in variable overtime.

II. Simple index number

Simple index number is that that in which base period is fixed.

III. Composite index number

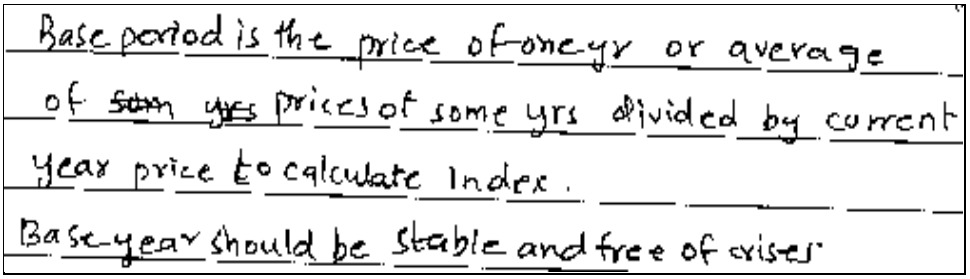
Composite index number is that in which bas year is considered from previous year

Question 6aii:

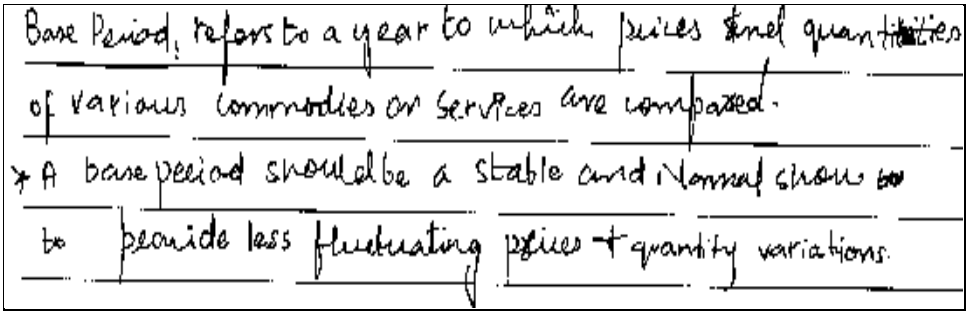
Define base period in the construction of an index number. Also write any ONE of its characteristics.

Better responses wrote the correct definition of base period of an index number. They were also clear about the characteristics of base period and were therefore, able to write a characteristic of the based period correctly.

Example 1:

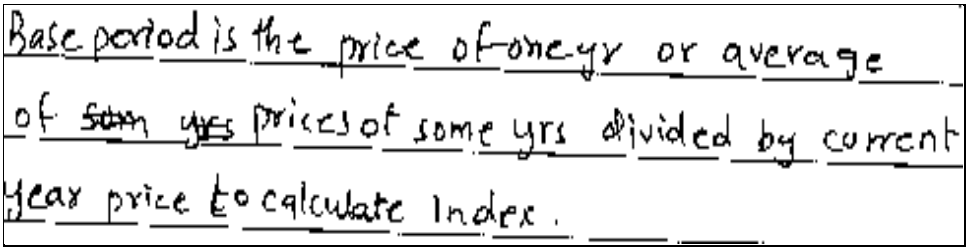

Base period is the price of one yr or average of ~~sum~~ yrs prices of some yrs divided by current year price to calculate Index.
Base year should be stable and free of crises.

Example 2:


Base Period, refers to a year to which prices and quantities of various commodities or services are compared.
* A base period should be a stable and Normal ~~chosen~~ to provide less fluctuating prices + quantity variations.

Weaker responses showed that the candidates unable to write the definition of base period and any of its characteristics correctly.

Example:


Base period is the price of one yr or average of ~~sum~~ yrs prices of some yrs divided by current year price to calculate Index.

Question 6b:

In the following table, the prices and quantities of three commodities in the years 2008 and 2009 are given. Construct the base year weighted index by taking 2008 as base year. Also comment on the result.

Commodities	Prices (Rs/ kg)		Quantities (kg)			
	2008	2009	2008	2009		
X	16	20	40	50		
Y	12	15	90	100		
Z	14	16	50	60		
Total						

Better responses indicated that the candidates had correct knowledge of base year weighted index number and wrote the formula correctly. From the given table, they found the values of $p_n q_o$, $p_o q_o$ and found the sum correctly. The total was substituted in the formula to find the correct answer.

Example :

Commodities	Prices (Rs/kg) P_o 2008	P_n 2009	Quantities (Kg) q_o 2008	q_n 2009	$P_n q_o$	$P_o q_o$
X	16	20	40	50	800	640
Y	12	15	90	100	1350	1080
Z	14	16	50	60	800	700
Total					2950	2420

Base year weighted index = Laspeyre's index.

Laspeyre's index = $\frac{\Sigma P_n q_o}{\Sigma P_o q_o} \times 100$ $\Sigma P_n q_o = 2950$ $\Sigma P_o q_o = 2402420$

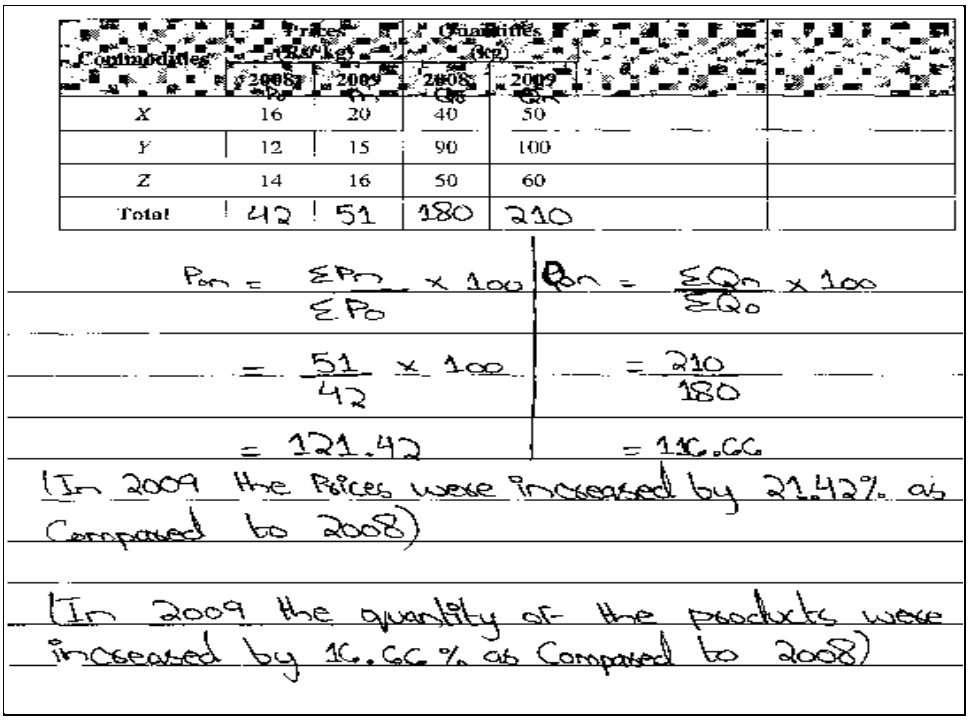
$= \frac{2950}{2420} \times 100 = 121.9\%$

The index gives a result of 121.9%. when computed keeping base year as 2008. This No shows that the prices have risen by 21.9% since 2008. As it this index keeps the base year quantities, the change in increase solely represents the change due to prices which is 21.9%.

:

Weaker responses displayed that most of the candidates were able to identify and label the four columns as p_o, q_o, p_n and q_n, but they could not figure out what was required in the last two columns. Use of incorrect formula was a frequent error too. Few more errors are evident from the given examples.

Example 1:



Commodities	Prices (Rs/kg)		Quantities (kg)		
	2008 P_o	2009 P_n	2008 Q_o	2009 Q_n	
X	16	20	40	50	
Y	12	15	90	100	
Z	14	16	50	60	
Total	42	51	180	210	

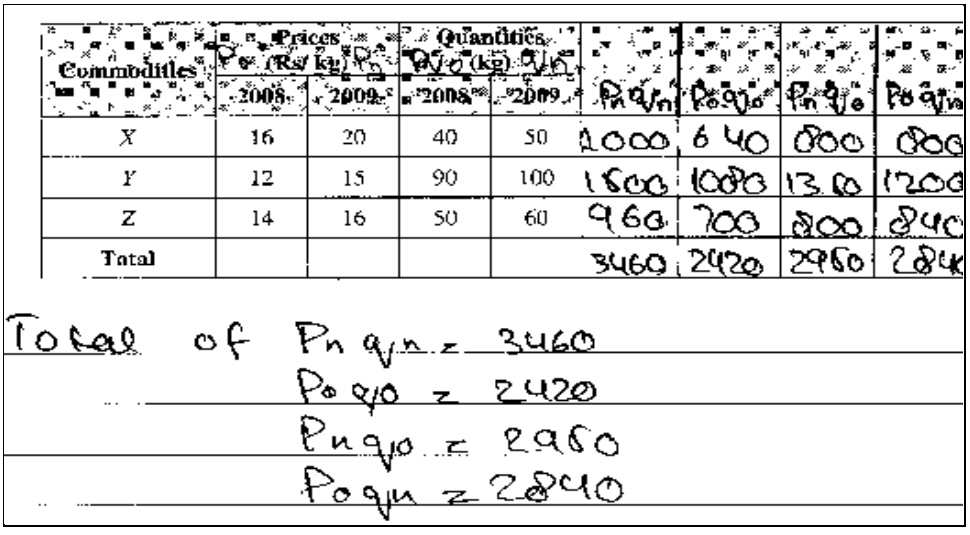
$P_{on} = \frac{\Sigma P_n}{\Sigma P_o} \times 100 = \frac{51}{42} \times 100 = 121.42$

$Q_n = \frac{\Sigma Q_n}{\Sigma Q_o} \times 100 = \frac{210}{180} = 116.66$

(In 2009 the Prices were increased by 21.42% as Compared to 2008)

(In 2009 the quantity of the products were increased by 16.66% as Compared to 2008)

Example 2:



Commodities	Prices (Rs/kg)		Quantities (kg)					
	2008 P_o	2009 P_n	2008 Q_o	2009 Q_n	$P_n q_n$	$P_o q_o$	$P_n q_o$	$P_o q_n$
X	16	20	40	50	1000	640	800	800
Y	12	15	90	100	1500	1080	1350	1200
Z	14	16	50	60	960	700	800	840
Total					3460	2420	2950	2840

Total of $P_n q_n$ = 3460

$P_o q_o$ = 2420

$P_n q_o$ = 2950

$P_o q_n$ = 2840

Question 7a:

ALGORITHMS is a ten-letter word.

i. How many passwords of 6 letters can be formed from the letters of the word ALGORITHMS, if

   I. each letter is used once?

   II. repetition of letters is allowed?

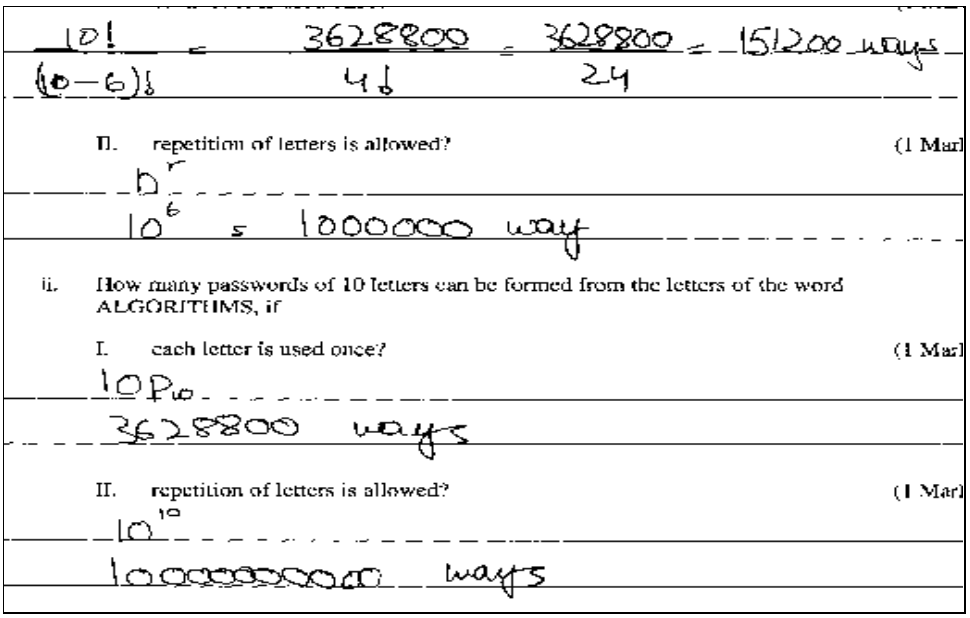
ii. How many passwords of 10 letters can be formed from the letters of the word ALGORITHMS, if

   I. each letter is used once?

   II. repetition of letters is allowed?

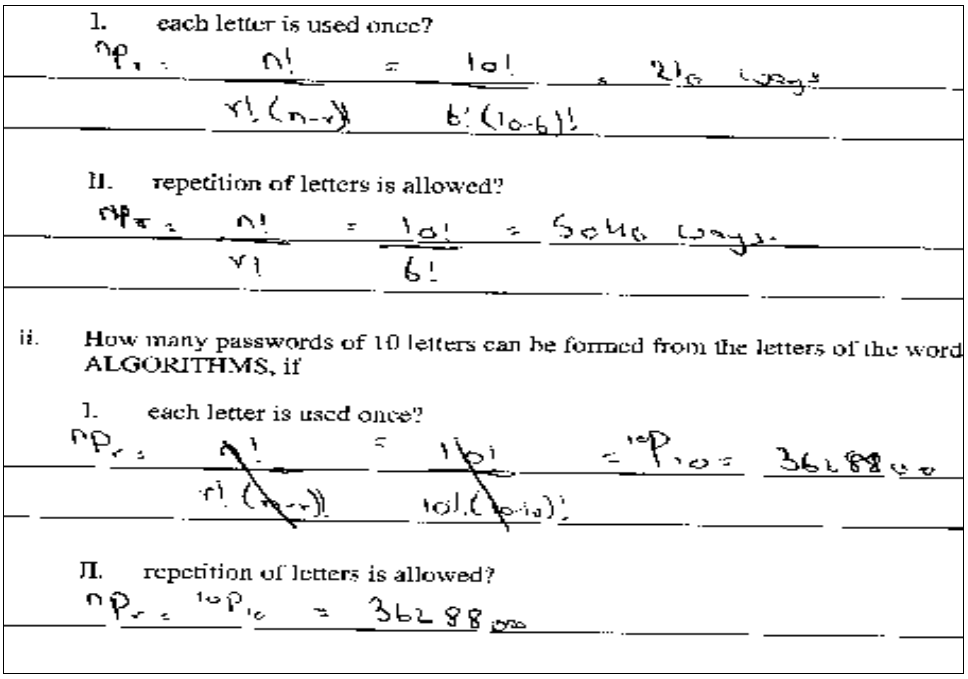
Better responses exhibited that for part **i**, candidates found the total number of passwords of 6 letters with and without repetition. In part **ii**, candidates were able to find the total number of passwords using 10 letters of the word ALGORITHMS. It showed that candidates have clear concepts of counting techniques and used appropriate formula to solve the given cases in the question.

Example:

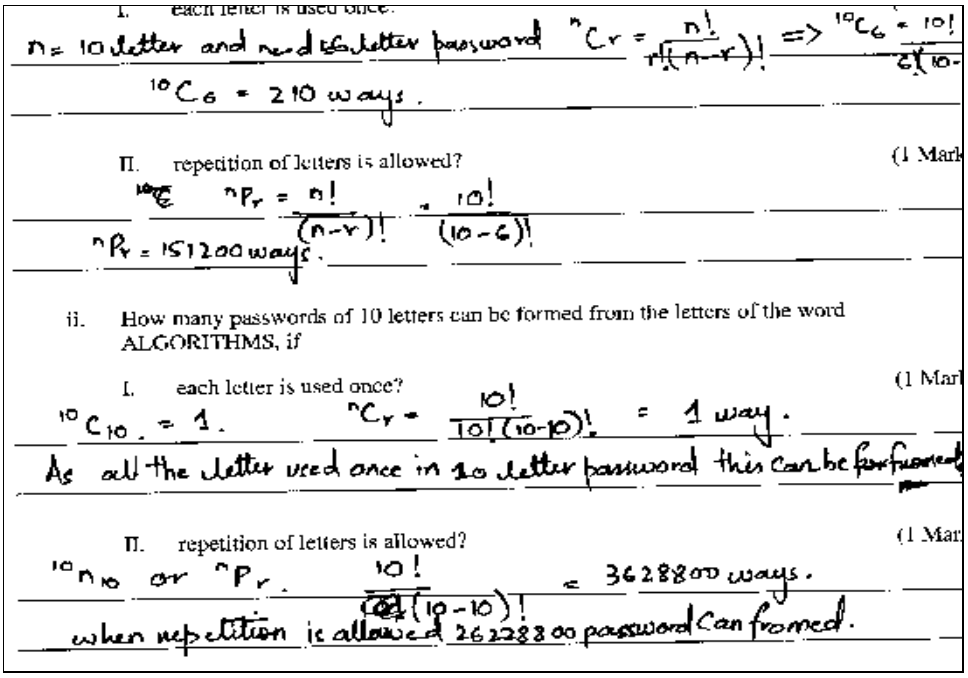


Weaker responses reflected ambiguities in the concepts of counting technique. Most of the candidates were failed to do part **i** and part **ii**. The most frequent mistakes noted is the wrong selection and application of formulae in different cases which is evident in the given examples.

Example 1:



Example 2:



Question 7b:

i. Write the sample space of an experiment when two coins are tossed and a die is rolled simultaneously.

(**Note:** Plural of die is dice.)

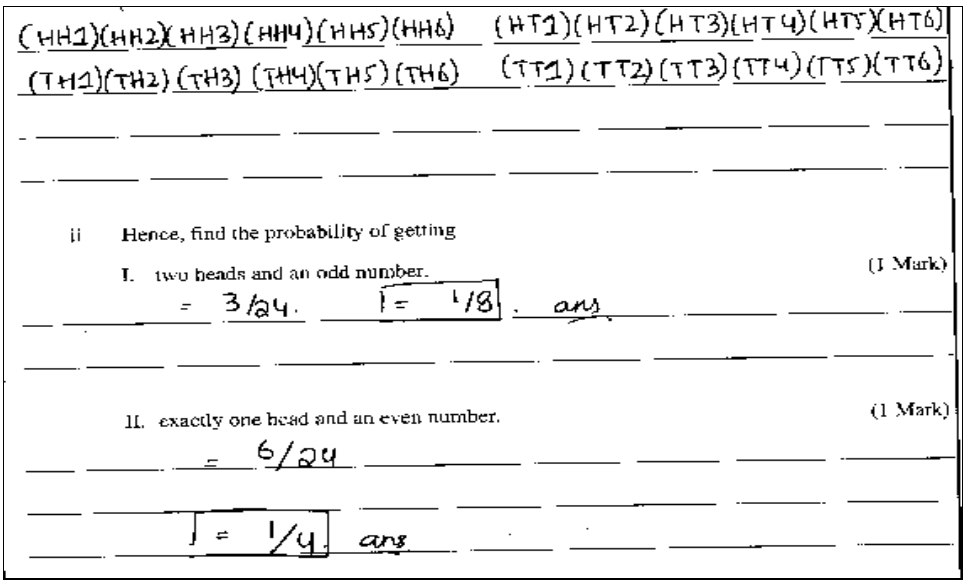
ii. Hence, find the probability of getting

I. two heads and an odd number.

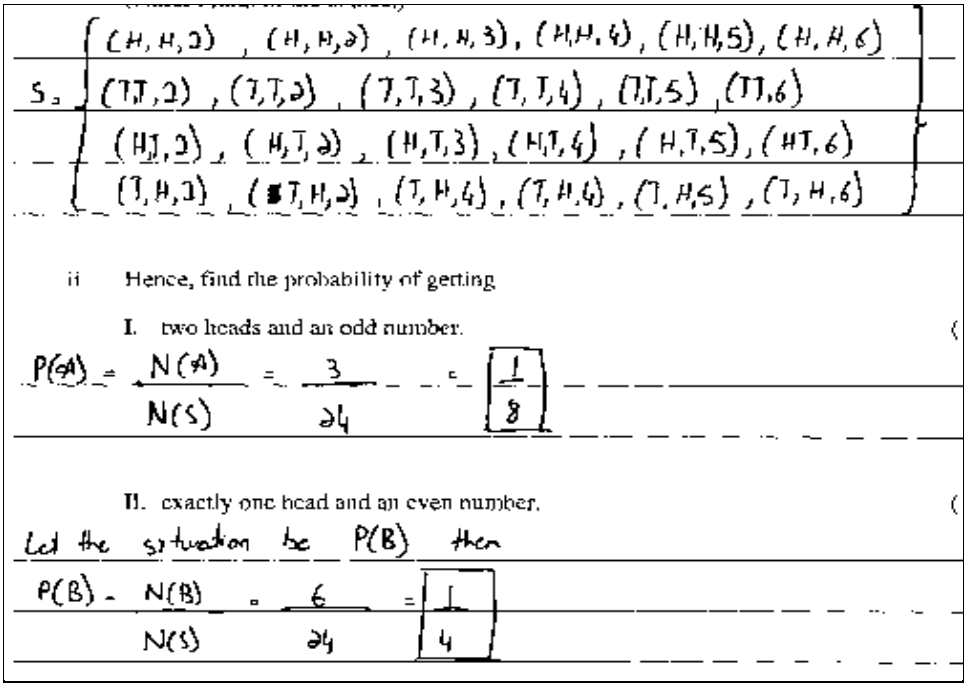
II. exactly one head and an even number.

Better responses indicated that the candidates were able to understand the given experiment and they were able to write the sample space correctly. Part **ii** was based on the sample space found in part **i** of the question to find the probability under different conditions. The candidates found the probability and translated the given conditions successfully to exhibit their command over the topic.

Example 1:

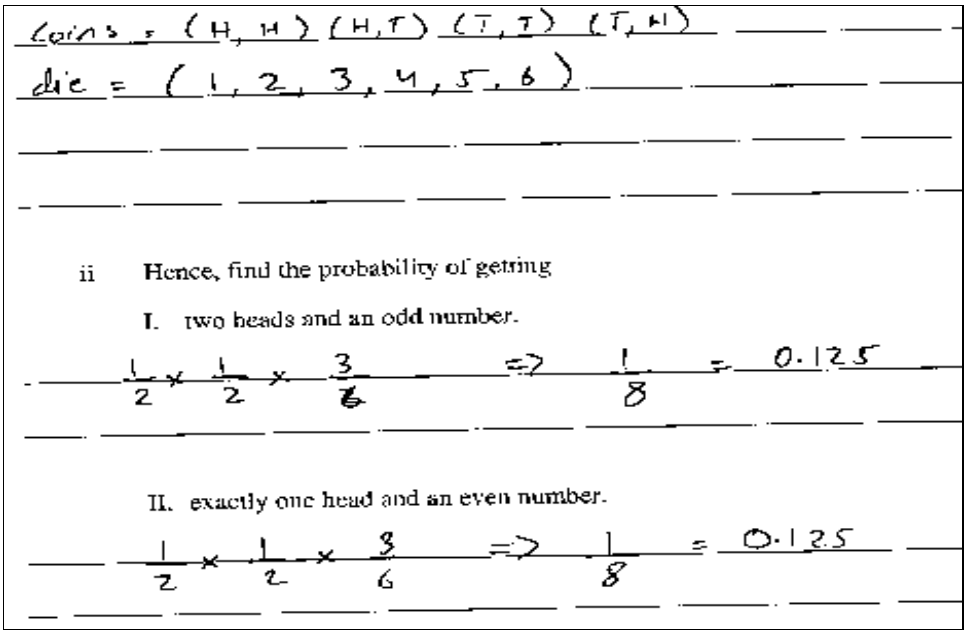


Example 2:

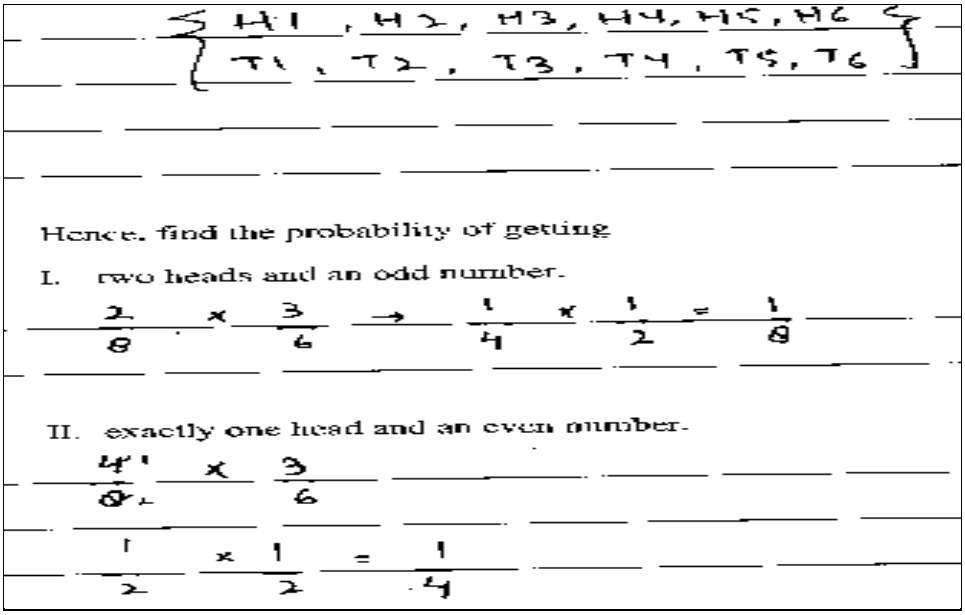


Weaker responses displayed that many candidates failed to understand the given situation or experiment. As a result, they were unable to write the sample space correctly and were failed to find the probability of the given cases based on the sample space found in the part **i** of the question.

Example 1:



Example 2:


{H1, H2, H3, H4, H5, H6
T1, T2, T3, T4, T5, T6}
Hence, find the probability of getting
I. two heads and an odd number.
2/8 x 3/6 → 1/4 x 1/2 = 1/8
II. exactly one head and an even number.
4/8 x 3/6
1/2 x 1/2 = 1/4